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A framework for soft mechanism driven robots

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Soft robots excel in safety and adaptability, yet their lack of structural integrity and dependency on open-curve movement paths restrict their dexterity. Conventional robots, albeit faster due to sturdy locomotion mechanisms, are typically less robust to physical impact. We introduce a multi-material design and printing framework that extends classical mechanism design to soft robotics, synergizing the strengths of soft and rigid materials while mitigating their respective limitations. Using a tool-changer equipped with multiple extruders, we blend thermoplastics of varying Shore hardness into monolithic systems. Our strategy emulates joint-like structures through biomimicry to achieve terrestrial trajectory control while inheriting the resilience of soft robots. We demonstrate the framework by 3D printing a legged soft robotic system, comparing different mechanism syntheses and material combinations, along with their resulting movement patterns and speeds. The integration of electronics and encoders provides reliable closed-loop control for the robot, enabling its operation across various terrains including sand, soil, and rock environments. This cost-effective framework offers an approach for creating 3D-printed soft robots employable in real-world environments.

Soft robots, characterized by their inherent elasticity and impact resistance, have emerged as highly adaptive and resilient technologies, particularly well-suited for challenging environments. Their effectiveness is evident in tasks such as navigating through debris fields[1] and overcoming steep obstacles[2]. Fabricated from a variety of materials including silicone elastomers, polyurethanes, and hydrogels, these robots demonstrate significant flexibility and compliance[3]. This not only facilitates safer interactions with humans but also enhances their capability to adapt to complex environments. Despite these advantages, the very flexibility that defines soft robots can also detract from their performance. Specifically, it can undermine their structural integrity and limit their range of movement, introducing challenges such as heightened friction and diminished speeds in comparison to their rigid counterparts[4–6]. This paradox underscores the need for ongoing research to balance the benefits of soft robotics with the inherent challenges posed by their material and design properties.

Pneumatic actuation, a common method for powering soft robots, allows for basic movements and crawling motions but limits speed and precision[5–9]. The crawling movements observed in pneumatic robots stem from the adoption of open-curve trajectories by their locomotion mechanisms. These trajectories involve movement patterns where the starting and ending points of an actuator are not contiguous, resulting in increased ground contact and reduced efficiency. Alternative actuation mechanisms like combustion and the use of shape memory alloys or dielectric elastomer actuators introduce unique capabilities for rapid movements and shape changes but face issues with controllability, power efficiency, and safety[10–14].

The operational scope of soft robots has historically been constrained by their dependence on tethers for power and control, leading to a growing interest in soft hybrid systems that integrate soft materials with untethered (rigid) electronic controls. Although soft robotic hybrids have shown promise, they have faced issues in power

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inefficiency and trajectory control[2,5,15–18]. Pneumatic soft robots with integrated fluidic circuitry, devoid of electronic components, have presented a promising avenue for enhancing resilience and autonomy in soft robotics, addressing constraints associated with tethered designs. However, limitations in computational abilities, wireless communication, and reduced locomotion speeds continue to constrain their practical use in real-world applications[19].

This discussion introduces a framework for the design and fabrication of soft hybrid robots. By employing multi-material printing and classical linkage design, we propose an easy-to-implement methodology that leverages the unique properties of soft materials while addressing their inherent limitations. Classical linkage design, in particular, facilitates the creation of closed-curve trajectories, enhancing power efficiency and movement precision without requiring actuators at every joint. This approach, inspired by the movement mechanisms found in legged animals, aims to emulate their efficient locomotion observed in nature. Four-bar linkages controlled by rotary actuators are adept at emulating terrestrial locomotion observed in bipeds[20], quadrupeds[21], and hexapods[22]. The majority of legged robots are composed of rigid materials[23,24] and are manually fabricated[25–29]; their implementations have lacked analyses of chassis and mechanisms designs for impact resistance, durability, and adaptability. DeMario et al.'s work similarly focused on multi-material printing of soft materials with varying Shore hardness to create locomotion mechanisms; the work, however, exhibits limitations. The study does not address key soft robotic attributes such as locomotion stability, impact resistance, and oscillation damping, which are essential for performance in varied environments. The use of a specific Klann mechanism with a single actuator and a miniaturized gear train significantly constrains the movement versatility of the robot. There is a limited investigation into material interfaces, with observed challenges like delamination during motion trials. The impact of variations in Shore hardness on locomotion is not explored. The use of PolyJet printing poses challenges regarding accessibility and cost, restricting its fabrication to specialized laboratory settings[30]. To further reflect upon the variations and capabilities of different printing methods for soft robotics, we created a comparison table in Supplementary Information - Comparison of Printing Technologies for Soft Robotics. We also compare the approach of using a single motor with a gear train to the approach of assigning individual motors to each leg in Supplementary Information - Comparison of Dependent and Independent Gait Control.

To merge the benefits of soft robotics with classical linkage design, we present a framework using multi-material fused deposition modeling (FDM). This approach involves the combination of soft and hard materials within unified geometries to achieve distinct compliance variations. This approach enables the creation of mechanisms that are both structurally sound and versatile in design, yet retain their inherent softness and impact resilience, addressing challenges noted in prior research. We systematically design, print, and assemble locomotion mechanisms and robot bodies from combinations of thermoplastic polyurethanes (TPU) of varying Shore hardness (75D, 95A, and 85A), with minimal human intervention. Our designed mechanisms feature joints with increased compliance relative to their links, enabling specific sections to bend under applied forces or moments. These flexible linkages are used as locomotion mechanisms in a soft quadruped (Fig. 1).

Our research undertakes fundamental challenges in soft robotics by directly tackling critical issues related to structural integrity and mobility. By integrating soft and rigid components with cost-effective fabrication techniques, we aim to broaden the capabilities and applications of soft robots. This approach addresses limitations in current actuation methods and improves interactions between robots and their environment, supporting the development of more versatile and power-efficient soft robotic systems (Supplementary Movie 1).

The goal of this framework is to facilitate the design and implementation of mechanism-driven soft robots suitable for real-world

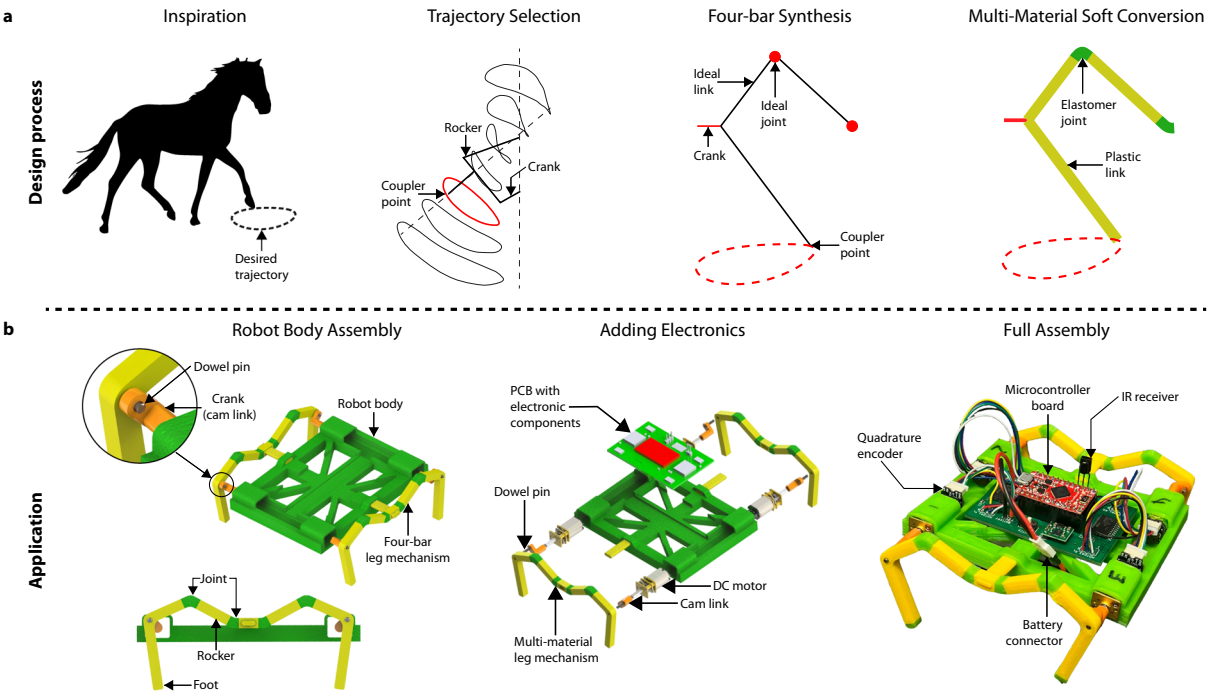


Fig. 1 | Framework extending classical mechanism design to soft robotics. **a** Design process. Inspired by nature, we observe the walking trajectories of quadrupeds such as horses. We then select a trajectory from a four-bar atlas. We synthesize the corresponding linkage in the ideal design domain. Finally, we convert the linkage into a multi-material design by replacing ideal joints and links with variations of soft and hard materials. **b** Application. We assemble the printed parts and integrate the electronics into the robot body, creating a `soft mechanism driven robot'.

environments, using low-cost materials and accessible desktop FDM printers.

Results

Material interfaces

Effective bonding between diverse materials is essential to enable mechanisms to withstand cyclic loading during operational use. Yin et al. investigated the interfacing properties between TPU and acrylonitrile butadiene styrene[31]; however, their study did not account for bending or elongation. In our initial experiments, we explored interfaces between polylactic acid (PLA) and TPUs, which led to delamination in our basic motion trials. In this study, we exclusively used TPUs; our approach mitigates the challenges associated with interfacing different materials by harmonizing the composition of our thermoplastics. We selected filaments with Shore hardness levels of 75D, 95A, and 85A due to their commercial availability and the balance between softness and structural integrity (Fig. 2). Filaments with lower Shore hardness are more prone to printing challenges and higher failure rates.

In single tool-head FDM printing, a material is deposited onto a layer that has recently been extruded and has not fully cooled, creating favorable fusion characteristics at that moment[32]. However, in a tool-changer setup, where materials are printed sequentially within a single layer, this process differs. Sequential printing of different materials increases the time between extrusions, allowing the previously extruded material to cool down. The fusion characteristics are less effective compared to single-material prints.

To address these challenges and understand the behavior of multi-material prints, we fabricated tensile testing specimens employing three distinct interfacing methods: straight, dovetail, and finger joints (Fig. 3a). We selected the dovetail and finger joint sizes and counts to maximize the contact area within the constraints of the specimen's size and the resolution of our printer. We also created specimens from uniform materials to assess strength degradation between single- and multi-material prints. We detail our tensile testing procedure in the Methods section.

Specimens with lower Shore hardness exhibited a capacity for higher strains, whereas specimens with higher Shore hardness demonstrated greater stress tolerance. The stress-strain behavior of two-material specimens is predominantly influenced by the properties of the material with a lower Shore hardness. Even in trials involving our firmest TPU (75D), which exhibited plastic behavior in uniform material trials, the overall behavior of the multi-material tensile testing specimens remained elastomeric. This observation is visible in both the stress-strain curves (Fig. 3c) and the Young's moduli (Fig. 3d–f) of our material combinations. This outcome underscores that combining materials with plastic behavior with those that display elastomeric properties results in a composite material that predominantly exhibits elastomeric behavior (Supplementary Movie 2).

In our experiments, straight interfaces, possessing the smallest contact surface area compared to finger and dovetail interfaces and lacking mechanical locking features, exhibited separation at relatively low-stress levels. Although a clear preference for dovetail or finger joints cannot be conclusively established due to their similar performance and the uncertainties inherent in our fabrication method, it was evident that their adhesive qualities surpassed those of straight interfaces (Supplementary Data File 1). Our finite element analysis revealed that the maximum stress exerted on the linkage is ~0.9 MPa during cyclic motion. This is notably lower than the tensile strength of the softest material we utilized (85A), which has a tensile strength of 4 MPa. Our model and experimental evaluation established that straight interfaces demonstrate sufficient interfacing efficacy for walking motions (safety factor ≥4). However, we note that dovetail or finger interfaces might be required in other specific robot applications involving greater force exertions. Supplementary Table 5 in the SI provides the ultimate tensile forces that each material and specimen combination can withstand before failure. This table is intended to serve as a reference for selecting appropriate materials and interfaces depending on the operation scenarios, creating new design selections for different robot applications. We also performed cyclic fatigue tests using ASTM standards. We show that all material combinations with all interfaces can endure a minimum of 10,000 cycles. The details of our

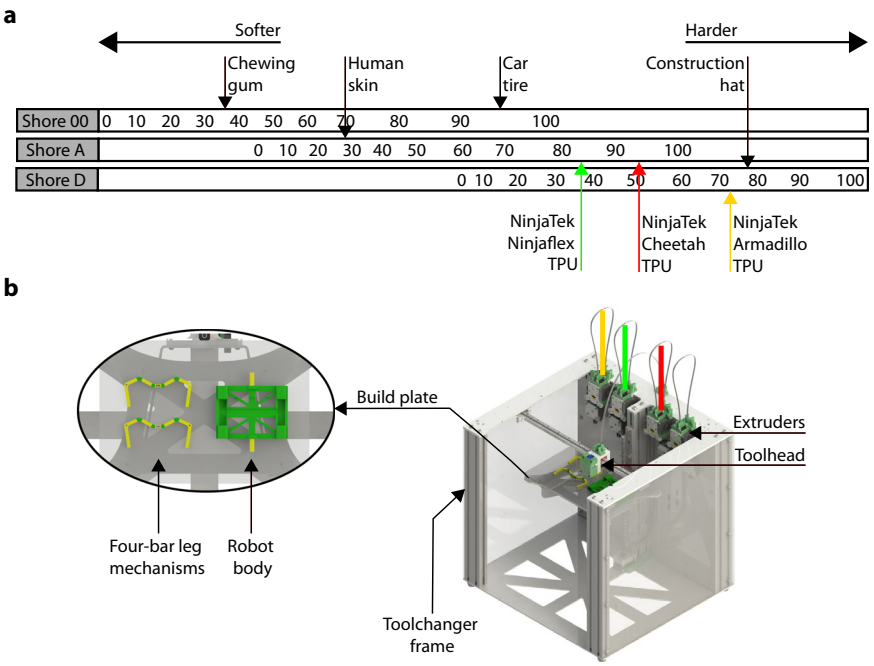


Fig. 2 | Materials and fabrication methodology. a The Shore hardness scale compares our filaments to other common materials. **b** The fabrication setup with three different filaments assigned to three separate extruders, and the resulting print of the leg mechanisms and the robot body on the print bed.

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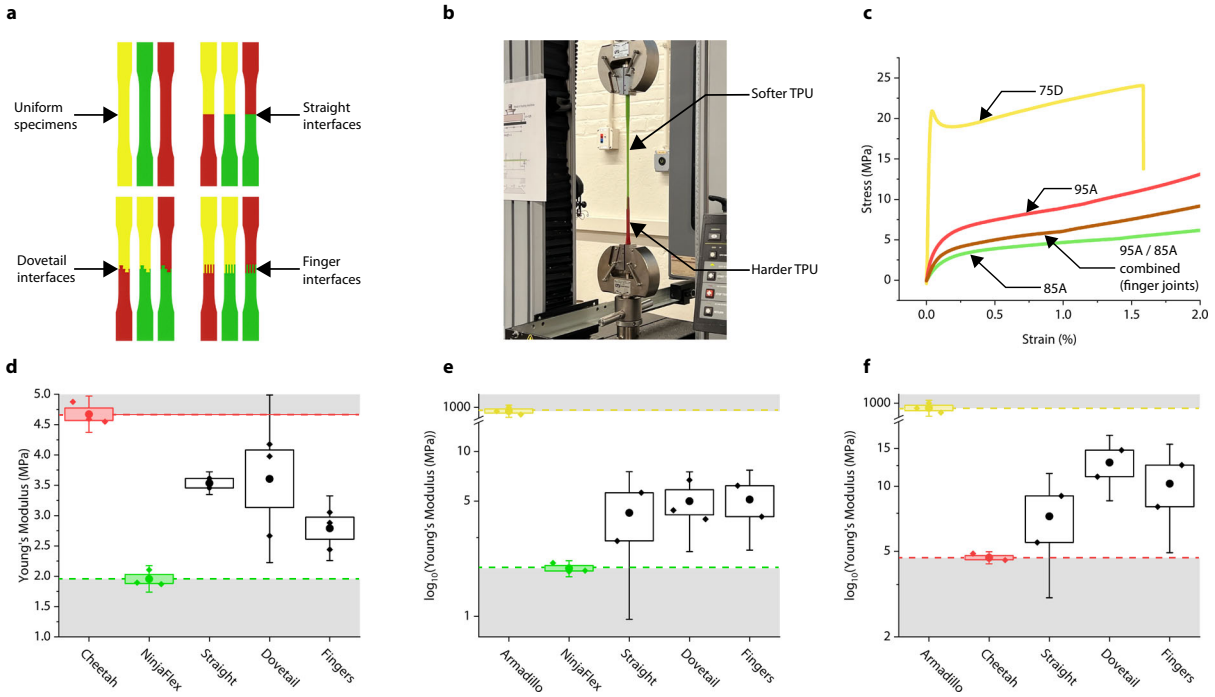


Fig. 3 | Tensile testing setup and Young's moduli experiments of multi-material tensile testing specimens. a The combinations of different specimens. **b** Test of a NinjaFlex (85A)/Cheetah (95A) specimen with finger interfaces. **c** A sample stress-strain graph comparing specimens of uniform Armadillo (75D), uniform Cheetah (95A), uniform NinjaFlex (85A), and Cheetah (95A)/NinjaFlex (85A) with finger interfaces. **d** Young's moduli of Cheetah (95A)/NinjaFlex (85A) specimens. **e** Young's moduli of Armadillo (75D)/NinjaFlex (85A) specimens. **f** Young's moduli of Armadillo (75D)/Cheetah (95A) specimens. Combined structures have Young's moduli ranging between the material variations.

tests are summarized in Supplementary Information - Cyclic Fatigue Testing. Our measurements are provided in Supplementary Data File 2. Published literature indicates that soft/hard multi-material specimens created with a PolyJet printer fail after ~1000 cycles[33], while our specimens withstand at least 10,000 cycles. We provide video snippets from fatigue tests in Supplementary Movie 3.

Quantitatively illustrating the stress-strain characteristics of multi-material specimens proves to be challenging. Unlike conventional engineering materials, elastomeric materials do not have a distinct yield point, which complicates the identification of the transition between elastic and plastic zones[34]. However, Young's modulus remains easily observable for both plastics and elastomers. We calculated and plotted the tensile modulus for each tested specimen, revealing that combined specimens consistently exhibit Young's moduli between their constituent materials (Fig. 3d–f). When comprised of an elastomeric and a rigid material, the modulus aligns more closely with the elastomeric material (Fig. 3d, e). When two elastomers are combined, the resulting modulus approximates the average of the two composing materials (Fig. 3f).

For the remainder of our experiments, we used straight interfaces and the material combination with the highest Shore hardness differential (75D/85A), which meets the requirement of $\gg$0.9 MPa for cyclic motion. The analysis in this section complements our overall framework by providing design guidelines for the development of other robots. For example, in applications where the robotic system must endure high stresses, finger or dovetail joints may be preferable to straight interfaces. Alternatively, in scenarios where impact resistance is critical, selecting the softest material combination over harder variants might be advantageous.

Multi-material mechanism driven soft-hybrid quadruped

We developed a four-bar locomotion mechanism for a terrestrial quadruped robot to showcase our methodology in a practical setting (Fig. 1a - Full assembly). Our linkage consists of a continuously rotating crank, a rocker that moves back and forth, and a third link that connects the two (Fig. 1b - Robot body assembly). In our Grashof crank-rocker mechanism, the crank, capable of continuous rotation and driven by a DC motor, actuates the entire linkage. The coupler can obtain an arbitrary shape if it connects the crank and rocker links. The design of the coupler determines the coupler points, which influence the overall trajectory of the leg. In our design, the tip of the coupler serves as the foot of the robot, creating a trajectory where the foot lifts at the rear, moves forward in the air, makes ground contact at the front, and then drags back, minimizing ground force exertion and hence friction. To understand how trajectory amplitude and range affect motion, we synthesized three linkages with varying trajectories. Please see Supplementary Data File 3 for the MATLAB scripts of the kinematic analyses of these linkages.

After the theoretical synthesis of the mechanisms, we translated the ideal joint-rigid pin structure into a multi-material print variation (Fig. 1a - Multi-material soft conversion). This adaptation was achieved by modulating compliance through the geometry of the linkage. In our design, areas designated for function as joints exhibit greater compliance compared to the links. This distinction was achieved by employing softer TPU materials (85A, 95A) for the joints and harder TPU materials (75D, 95A) for the links, strategically varying material hardness to meet the specific functional needs of each component. We performed deflection simulations using finite element analysis (FEA) to compare the displacement differences among three material combinations. We provide a detailed analysis in Supplementary Information - Deflection Analysis and the corresponding simulation file in Supplementary Data File 4. We provide an analysis of the power consumption and efficiency of each mechanism in Supplementary Information - Mechanism Efficiency Testing.

We fabricated three distinct configurations: links made of 75D TPU with joints of 85A TPU, links made of 75D TPU with joints of 95A

TPU, and links made of 95A TPU with joints of 85A TPU (Fig. 2a). The tuned fabrication profile with the entire printer configuration is shared in Supplementary Data File 5. The crank, serving as the connector between the motor and the rest of the linkage and undergoing a full 360° rotation, is unsuitable for a compliance-based multi-material design due to its functional demands. Therefore, we 3D printed it separately using stereolithography (Prusa Tough resin) and subsequently attached it to the linkage with a dowel pin.

In traditional four-bar mechanisms, an offset is necessary at the joint locations to accommodate the pin connecting the links, preventing a completely planar link design. However, our fabrication technique, which facilitates the differentiation of materials within the same print layer, eliminates this requirement. We can 3D print the leg mechanisms in a streamlined, single-click process using our multi-material tool-changer system. The limitation is not completely diminished but reduced to the only exception of the cams that connect the motors to the linkage (Fig. 1). The cams are fabricated separately and still need an offset; this approach not only simplifies the manufacturing process but also enhances the design efficiency of the leg mechanisms.

The body of the robot is designed as a monolithic system, comprising the main frame made from a softer variant of TPU (85A) to ensure flexibility. The body is complemented by connection beams for the legs, fabricated from a sturdier TPU variant (75D) to improve structural integrity. This combination of materials optimizes the overall performance of the robot by balancing flexibility and strength in its construction. Although changes in body design can influence robot behavior, this work focuses specifically on the leg mechanisms that drive the robot. Given that the robot body offers a vast design space, we kept the robot body mostly constant throughout this work to control the experimentation of the leg mechanisms. The only exceptions are two test series. In Supplementary Information - Body Material Testing, we varied the Shore hardness of the robot body while keeping the leg mechanism unchanged. In Supplementary Information - Body Geometry Analysis, we varied the geometry of the robot body while keeping the leg mechanism unchanged.

We attached the legs to the robot body using the connection beams. The computer aided design (CAD) files of the legs and the robot body are provided in Supplementary Data File 6. The soft body of the robot holds four DC motors, four magnetic encoders, and a custom-made Printed Circuit Board (Supplementary Data File 7) that interconnects the electronic components (Fig. 1b - Adding electronics). We implemented closed-loop controllers for the motors to achieve our walking gait (Supplementary Data File 8).

Trajectory analyses

In designing the locomotion mechanism, our main objective was to achieve a trajectory resembling a reverse D-shape (Fig. 1a - Inspiration). The flat portion of the D represents the supporting phase, where the foot touches and drags along the ground. Maintaining flatness during this phase is crucial as it minimizes the application of force on the ground, reducing normal forces and friction. The remaining arc depicts the lift-off and forward motion of the foot through the air.

We derived analytical expressions for linkage motion primarily through kinematic analysis, ensuring that the linkages accurately follow the desired trajectories essential for fast and effective movement. By developing kinematic equations for a standard four-bar linkage using trigonometric identities and vector loop equations, detailed in the Supplementary Fig. 1, our approach enables us to determine the position of the linkage's endpoint (i.e., the foot) based on specific rotary inputs to the crank (Fig. 4a). We included an analytical model for the rigid versions of our four-bar linkages, along with a MATLAB script for easier analysis, in Supplementary Data File 3.

However, this analysis does not fully capture the motion dynamics of the soft mechanism, as the joints in our multi-material configuration exhibit resistance to motion due to their spring constants and damping effects, unlike ideal joints. We conducted two additional analyses: image trajectory tracking (Supplementary Data File 9) and finite element analysis (Supplementary Data File 10), to compare the trajectory of our linkage with the theoretical (rigid) model. We provide the data from all these experiments in Supplementary Data File 11. Supplementary Movie 4 features a video from our COMSOL simulation, and Supplementary Movie 5 shows the image tracking procedure.

Our findings indicate that the trajectories produced by our mechanisms closely align with their theoretical counterparts in terms of shape (Fig. 4d, e). The trajectories derived from video tracking (Fig. 4d) and finite element analysis (Fig. 4e) are consistent in both shape and size, with an exception for models using stiffer (95A) joints. In our video tracking experiments, we observed that despite immobilizing the motors and body, the stiffer joints introduced resistance that altered the position of the motor and angled the body, significantly deviating the trajectory from the expected outcome. This discrepancy was not observed in finite element analysis, where, due to simplifications made to reduce computational complexity, such physical distortions were not accounted for. We provide the corresponding simulation file in Supplementary Data File 10 and the simulation video in Supplementary Movie 4. The layered nature of 3D printed systems, material flow inconsistencies, and external factors such as humidity further contributed to fabrication-related deviations.

Our observations reveal that increasing joint stiffness leads to a reduced horizontal range in trajectories. The softness of the links appears to have minimal impact on the horizontal range. This suggests that while most deformation occurs at the joints, the links also undergo a slight elastic deformation. However, this deformation is relatively minor compared to that of the joints, affecting the horizontal range of a trajectory only by 2%. We also noticed that our motion curves are smooth, a characteristic we attribute to the damping effects inherent in the soft joint structures. We also performed a mean error analysis to numerically compare the trajectories. We provide details of our analysis in Supplementary Information - Mean Error Analysis for Air Trajectories and Supplementary Data File 12.

Locomotion tests

To examine the impact of varying compliance (i.e., multi-material combinations) in leg mechanisms, we conducted locomotion tests. Our robot, equipped with four independent actuators and encoders, allowed for individual closed-loop control of each leg mechanism, enabling the execution of any quadruped gait. We implemented the trot gait on the quadruped as it offers a trade-off between speed and stability. In the trot gait, two diagonal legs operate in synchrony, while the other diagonal pair is offset by 180° [35].

In the future, sensory feedback from sources such as cameras could enable adaptive gait changes, allowing the robot to change gaits in response to various obstacles. To prioritize ease of fabrication, reduce costs, and maintain focus on developing mechanism-based soft robots, we conducted all of our experiments using the trot gait.

For a controlled comparison, we maintained a consistent robot body while testing five distinct leg mechanisms. This set comprised three variations of our synthesized main mechanism (Fig. 5d, designs 2i, 2ii, 2iii), and two additional mechanisms designed for wider and taller trajectories, constructed from Armadillo (75D) and NinjaFlex (85A) materials (Fig. 5c, designs 1 and 3). To track the foot and geometric centroid of each robot during locomotion, we employed an image trajectory processing algorithm shared in Supplementary Data File 9.

The results show that the combinations of materials directly affect the smoothness of locomotion (Fig. 5). We provide our data and videos of the experiments in Supplementary Data File 13 and Supplementary Data File 14. Using combinations of softer materials dampens the amplitude of the vertical movement of both the foot and the centroid

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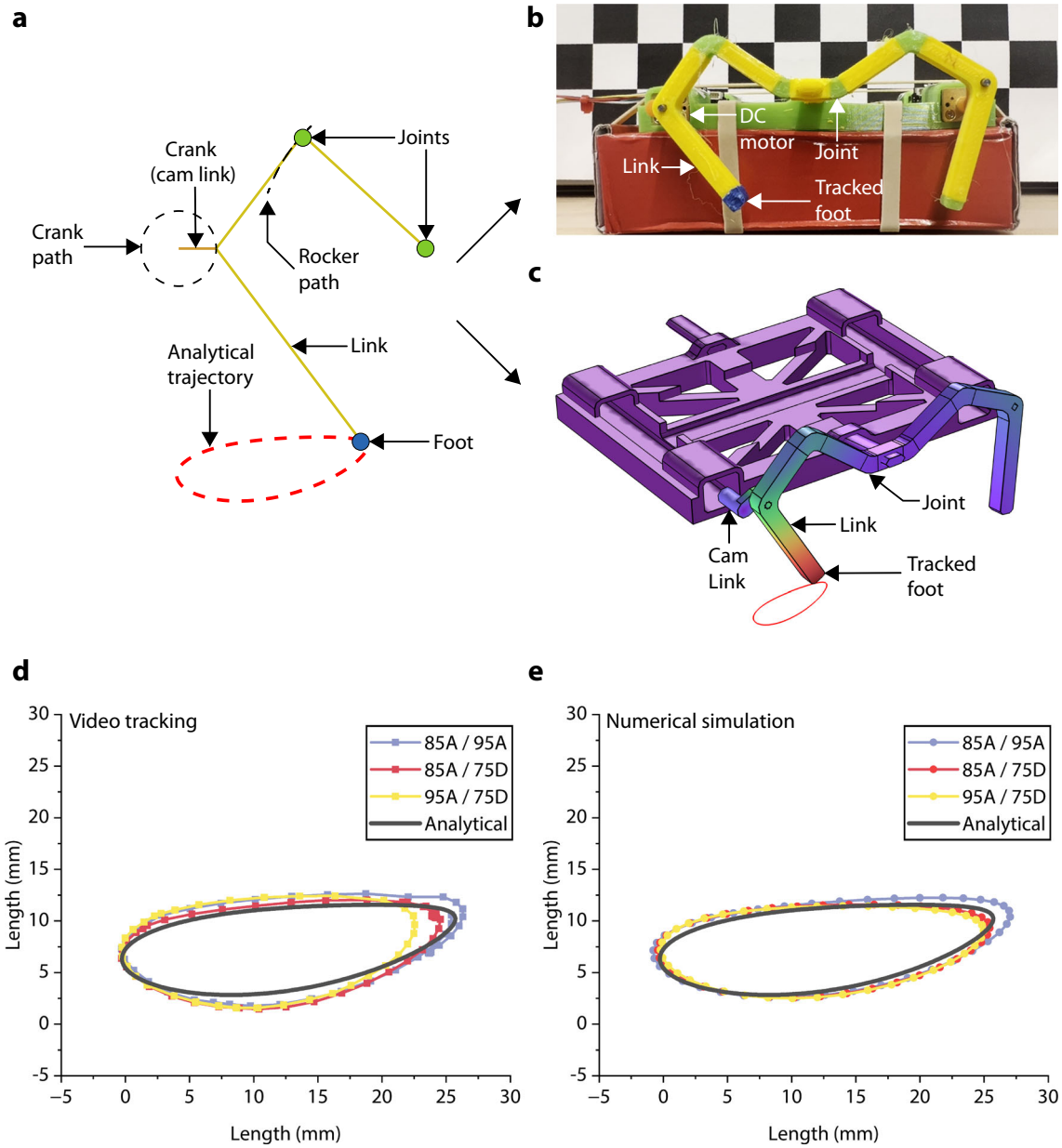


Fig. 4 | Trajectory analysis. a Analytical synthesis of the mechanism generating the desired trajectory. **b** The leg mechanism being actuated in mid-air to track the foot for trajectory generation. **c** Monolithic mechanism duplicated in FEA domain numerically calculates the foot trajectory. **d** Trajectory data acquired with image processing. **e** Trajectory data generated in finite element simulations.

of the robot (Table 1). Softer materials, compared to rigid materials, undergo greater elastic deformation under the same applied force, effectively acting as dampers to attenuate motion roughness. We also observed that variations in analytical synthesis translate well into the fabrication domain, with wider and taller trajectories apparent with different geometrical leg variations (Table 1, Fig. 5). In summary, increased material softness enhances speed while also reducing the vertical range of robot movement (Table 1).

Although the primary focus of the paper is on soft locomotion mechanisms and the corresponding locomotion tests primarily explore their properties, we recognize the significant influence of body stiffness on motion dynamics. To examine this effect, we performed additional locomotion tests, keeping the locomotion mechanism constant while varying the body material (85A, 95A, and 75D). Detailed descriptions of these tests and the associated data are available in Supplementary Information - Body Material Testing and Supplementary Data File 15. The results from these tests demonstrate that softer robot bodies exhibit increased undulations, smoother motion, reduced centroid oscillations compared to non-elastomeric bodies, and, consequently, more stable locomotion. Supplementary Movie 6 provides video footage of these experiments. A robot body made from 85A TPU exhibited greater angulations than a robot body made from 95A TPU, resulting in higher amplitudes in vertical movement. However, angulations did not relate to rough motion. Instead, the increased angulations suggest improved energy dissipation through body movement, resulting in smoother locomotion (Supplementary Movie 6).

As a final analysis, we explored how geometric differentiation, in terms of cut-out and beam usage, affects the deformation and angulation capabilities of the body. We conducted a finite element simulation comparing three different body models. Our findings indicate that bigger cutouts enable bigger angulation capabilities, but the

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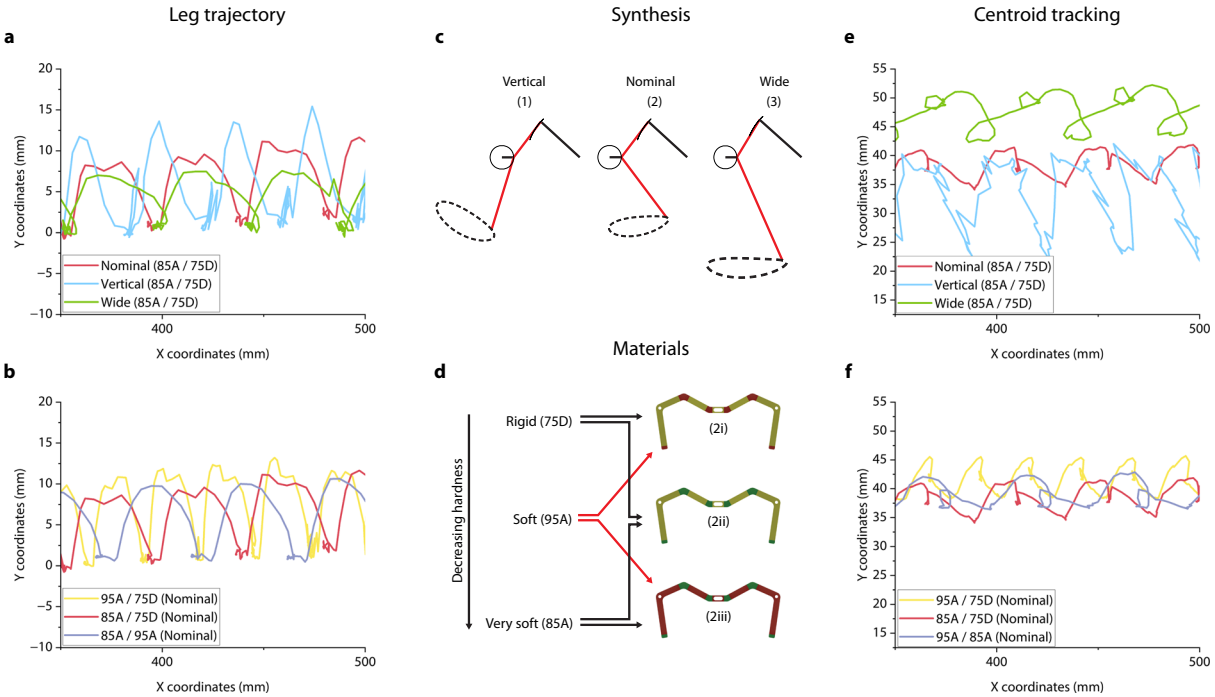


Fig. 5 | Experimental data on the effects of different linkages or material combinations on locomotion and centroid movement. Material combinations exert a discernible impact on motion smoothness, with softer combinations correlating with diminished oscillations and expanded range in locomotion direction. **a** Image tracking of leg trajectory for varying synthesis combinations. **b** Image tracking of leg trajectory for varying material combinations. **c** Different synthesized mechanisms tested. **d** Different material combinations tested. **e** Centroid image tracking for varying synthesis combinations. **f** Centroid image tracking for varying material combinations.

analysis does not capture information regarding the structural integrity and load-carrying capacities of robot bodies. Our interpretation is that a balance between supports and reliefs is required to establish a successful operation, and this balance may shift with the operation and design needs. The details of these tests are shared in Supplementary Information - Body Geometry Analysis, and the simulation files are shared in Supplementary Data File 17.

Impact resistance

Given that our robot chassis and locomotion mechanisms were constructed from a combination of hard and soft materials within unified geometries, our aim was to demonstrate that these overall geometries retained soft characteristics regarding impact resistance and elastic deformation. We devised a hard version of our robot comprising rigid links and conventional joints connected via dowel pins, akin to a standard classical mechanism. We subjected our soft robot and its hard counterpart to compression tests for comparative analysis. We share our test setup in the Methods section and the data in Supplementary Data File 16. We emphasize that this rigid robot was used solely for impact resistance tests, as design and component modifications would have been necessary to achieve locomotion with rigid materials.

The results presented in Fig. 6 demonstrate that the integration of soft and hard materials in our soft hybrid robot exhibits overall soft behavior. We provide videos of our tests in Supplementary Movie 7. Deformations primarily occur within the soft sections of the body and locomotion mechanisms. In contrast, the hard variant experienced irreversible plastic deformation and fracture. These findings align with our tensile testing results, indicating that monolithic geometries composed of soft and hard materials collectively exhibit soft behavior, with the structural elements of our robot working similarly to those of a soft robot.

Table 1 | Comparison of speeds and step ranges of the robot for different combinations of materials for legs

Variation (trajectory/joints/links)	Speed (body-length/s)	Horizontal range (mm)	Vertical range (mm)
Nominal/95A/75D	0.154	26.89	11.55
Nominal/85A/75D	0.419	43.14	9.09
Nominal/85A/95A	0.428	46.97	9.29
Vertical/85A/75D	0.354	26.84	13.68
Wide/85A/75D	0.364	46.66	7.10

Demonstrations

Following experimentation, we tested our quadruped in different real-life scenarios (Fig. 7, Supplementary Movie 1). We operated the robot on different terrains, including soil with gravel (Fig. 7a), rocks (Fig. 7b), dirt (Fig. 7e), sand (Fig. 7f), and carpet (Fig. 7g). Our locomotion mechanisms enable the robot to achieve different trajectories, and the soft angulation capabilities of the entire robot structure show adaptability in navigating diverse terrains. We demonstrate that variations in the synthesis of locomotion mechanisms are key to adapting to different mission scenarios. For instance, our vertical step linkage (Fig. 5c, design 1) successfully climbed a steep slope (Fig. 7d), a task that our standard linkage (Fig. 5c, design 2) was unable to accomplish. Finally, we show the durability of our structural elements, both the robot body and the locomotion mechanisms, by driving over the robot chassis with a car (Fig. 7c and Supplementary Movie 8).

Across various demonstrations, the angulation of the body becomes notably evident. This observation is crucial as it highlights how the movement of the body mirrors that of biological systems, with oscillations counteracting the forces generated by the robot's

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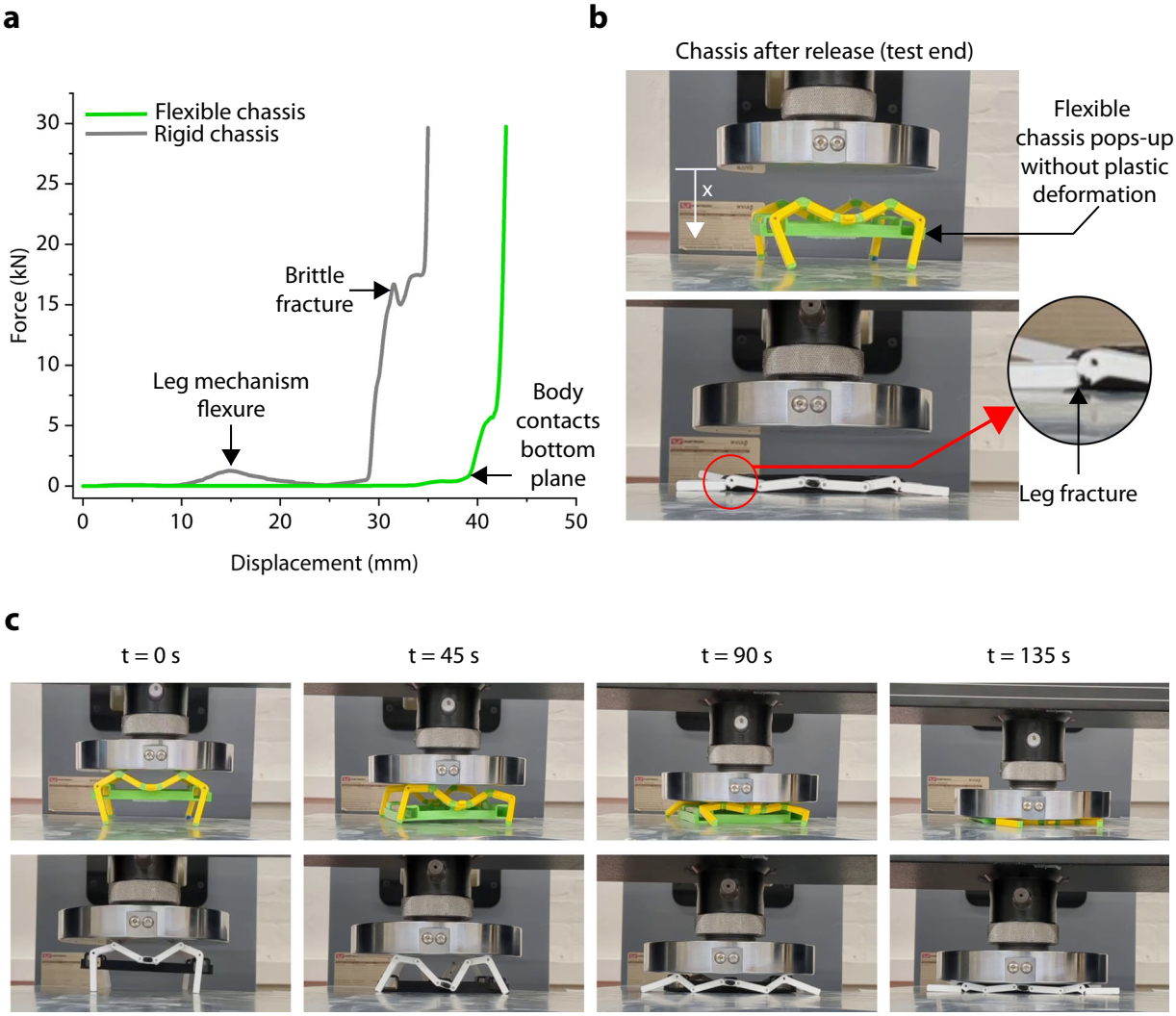


Fig. 6 | Impact resistance testing on the flexible and rigid robot chassis. a The force-versus-displacement graph from the impact test illustrates that the flexible chassis endures deformation before forces escalate. Spikes in the rigid chassis plot denote fracture points. **b** Post-test, the flexible chassis reverts to its original shape, while the rigid chassis remains flattened due to plastic deformation and fracture. **c** Picture of the test setup.



Fig. 7 | Robots operating in different environments. a Robot walking on soil. **b** Robot walking on a rock. **c** Robot driven over by a car. **d** Robot climbing steep rock (vertical trajectory - Fig. 5c). **e** Robot walking in dirt. **f** Robot walking on sand. **g** Robot walking on a carpet.

locomotion, thereby significantly improving stability (Fig. 7g). We describe stability by a combination of different properties such as the vertical amplitude of centroid movement, slope of the curve of the movement plot, and the tip of the plot (being sharp or plateu-like). Elastomeric leg mechanisms and robot bodies exhibit smaller amplitudes in centroid movement; however, comparing two different elastomeric bodies reveals that the softer material may display greater movement due to enhanced angulations. These angulations suggest an

increase in energy dissipation through body movements, leading to more stable locomotion. This represents a significant advantage of our fabrication framework over rigid quadrupeds, which face higher normal forces and consequently increased oscillations.

To demonstrate the effectiveness of our framework in real-world scenarios, we tested the maximum distance our robot could travel on a single battery charge. For this purpose, we changed the initial battery (3.7 V, 750 mAh) with a battery that still fit the form factor of the robot but had a higher discharge capacity (2000 mAh). We programmed our fastest locomotion mechanism, which featured a nominal trajectory with 95A links and 85A joints (Table 1). The robot, with a body length of 98.46 mm, covered an untethered distance of ~250 m (or 2500 times its body length), indicating the practical potential of our framework. As our cyclic fatigue tests showed that our interfaces can withstand at least 10,000 cycles; we deduct that the operational range of the robot is constrained by its battery capacity.

Discussion

The field of soft robotics has emerged as a transformative avenue in the realm of robotics, offering consequential advantages in adaptability and impact resistance. Through the synthesis of compliant materials and innovative design principles, a soft robot not only mimics the dexterity and flexibility of biological systems but also competes with a conventional rigid robot in its ability to navigate complex and dynamic environments (Fig. 7).

Although soft robotics offers a myriad of advantages, a notable struggle within the field lies in the domain of terrestrial locomotion. The very softness of the materials that endow these robots with unique capabilities also presents challenges when it comes to achieving efficient and precise movements on solid ground. The inherent compliance and deformability of soft materials, which contribute to adaptability and impact resistance, can impede traditional methods of generating controlled and stable locomotion. The lack of rigid structural elements, which is prevalent in conventional robotics (and vertebrates), poses difficulties in maintaining the necessary stability and directional control essential for terrestrial mobility.

Our design and fabrication framework confers a distinct advantage over numerous existing soft robots: practical suitability for real-world scenarios and the preservation of inherent softness while maintaining structural integrity. By facilitating the development of robots capable of adapting to various terrains and trajectory specifications, the field is positioned to introduce soft hybrid robots into practical use. These robots seamlessly transition from the typical crawling motion observed in many soft robots to walking via closed-curve trajectories. The potential customization of these robots with a variety of sensors to fulfill specific application requirements opens up diverse avenues for applications and future research.

Our framework brings classical mechanism design principles into the soft robotics domain; enabling flexible, scenario-specific designs. This approach allows for the replication and conversion of any other mechanism, such as the Watt linkage, which converts rotary to linear motion, into the soft robotics domain. Our goal is to provide the robotics community with a practical method to design and build mechanisms that help soft robots meet specific operational needs.

Our research has also identified key limitations, particularly in integrating conventional electronics and circuits with our robot design. Although structural elements are robust, capable of carrying loads and accommodating angular distortions, the inclusion of rigid electronic components restricts these capabilities. Future work will incorporate flexible electronics and encase electronic components entirely within soft-mechanism-driven robots. By using soft materials to absorb the forces that act on the robot body, fragile electronic components will be protected from stresses. To ensure consistency and eliminate potential errors associated with manual operation, we will explore the opportunity of integrating new tool heads, such as pick-and-place and solder dispensing, with a tool changer, and automate the integration of electronics into the robot chassis.

Printable TPUs and their material properties, such as glass transition and heat deflection temperatures[36], impose constraints on the types of robots that can be produced using our framework. For example, robots developed with this approach may quickly become inoperative in extreme temperature environments.

The locomotion range and speed of the robots could be further enhanced by incorporating high-performance miniature motors. Waterproofing electronics, including motors, would expand their applicability to water surface and subsurface environments.

It is important to note that transitioning traditional mechanisms to an entirely compliant basis is not feasible for continuous rotation, as our robot requires distinct rigid cam links for the continuous rotation of its legs. Although alternatives, such as linear actuators to avoid complete rotations, are possible, design constraints will still remain within the system.

Methods

Tensile testing

We tested the specimens using an Instron 5567A Universal Testing System (Fig. 3). We designed the specimens in accordance with ASTM D638 Standard and extended them with a 500 mm min^{-1} rate until they failed or the range of motion of the testing system was reached. We tested the three interfacing methods (straight, dovetail, and fingers) with the combinations of three filaments (85A, 95A, and 75D), leading to nine different combinations for multi-material prints with an additional three single-material prints for comparison. We printed and tested each combination three times.

Upon initial examination, we identified outliers within our data, characterized by premature failure points in contrast to other specimens. Despite using identical materials and print parameters for similar combinations, inconsistencies emerged due to low resolution in layer height, fluctuations in idler tensioning, wear and loosening within the system leading to minor alignment offsets, humidity (given the hygroscopic nature of TPU), and filament path obstructions. Following the removal of outlier data, we obtained meaningful information to facilitate a comparison of interface characteristics.

Image processing analyses

We recorded videos of the mechanisms during actuation in mid-air. We marked points of interest for tracking with bright blue tape to ensure contrast against the mechanisms and background (Fig. 4b). These videos were imported into MATLAB, where they were segmented into individual frames for analysis. For each frame of the recorded videos, we captured the RGB value of each pixel. A color thresholding mask was applied to isolate the pixels of interest at the foot of the mechanism. To consolidate the points, we computed their centroids at each time step. This process was replicated for all frames in the video, enabling us to plot the trajectories. This procedure was performed for our nominal four-bar linkage (Fig. 5), with joint differentiation. We tested linkages with joints from TPU with 85A and 95A Shore hardness and links from TPU with 75D Shore hardness to understand the effect of hardness variation on motion.

Finite element analyses

We created a CAD model assembly of the linkage, the body, the cam (which couples the DC motor to the linkage), and a dowel pin connecting the cam to the linkage. This model was imported into COMSOL Multiphysics in standard for the exchange of product data format, in which we created an analysis with the multibody dynamics tool (Fig. 4c). The multibody dynamics tool allows different bodies to be meshed separately, creating an assembly rather than a unified solid body. We created separate entities of the linkage, the cam, and the pin and connected them with joint definitions.

In our model, we specified the multi-material structure by developing material models for the different TPU filaments used in fabrication and assigning them to the appropriate sections of the linkage. We conducted simulations on three different material combinations: 85A TPU joints with 95A TPU links, 85A TPU joints with 75D TPU links, and 95A TPU joints with 75D TPU links. In each simulation, we applied a constant rotary motion (speed of π rad s^{-1}) to the model at the hinge joint connecting the cam to the motor.

Compression testing

To assess the impact resistance of the robot, we performed compression tests using a universal testing machine (Fig. 6). For comparison, we fabricated an entirely rigid version of the robot from PLA (Shore hardness 98D[37]). This rigid model featured rigid links and ideal pin joints connected using 3 mm steel dowels. In both models, electronics and motors were excluded during tests. We performed our tests with an Instron 5567A Universal Testing System, equipped with compression plates and a load cell. Each robot chassis, positioned upright, underwent compression at 30 mm min^{-1} until structural failure occurred or the lower limit of the Instron was reached.

The soft chassis was able to withstand a compressive force of ~30 kN after being completely flattened, undergoing only elastic deformation. In comparison, the legs of the rigid chassis fractured after enduring ~17 kN. The body began to deform plastically at ~18 kN. Upon release of the compression platens, the rigid chassis remained flattened with non-reversible damage to the body and near-complete fracture in the legs, whereas the soft mechanism returned identically to its initial configuration and sustained no lasting damage.

AI usage

We used generative AI (ChatGPT, 4o) to generate the Python code in the "Mean Error Analysis for Air Trajectories" section in Supplementary Information. We described the structure of the mathematical method ourselves, and AI created the code block accordingly.

Data availability

All data necessary to evaluate the conclusions in the paper are available in the paper or the Supplementary Information.

Code availability

All code necessary to evaluate the conclusions in the paper are available in the paper or the Supplementary Information.

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Author contributions

C.A. designed the research and wrote the paper. C.A. and B.J.K. developed the fabrication pipeline. C.A., C.G., and S.A.F. performed experiments. C.A. and C.G. developed the control algorithm. M.P.N. supervised the work and edited the paper.

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Competing interests

The authors declare no competing interests.

Additional information

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